space structures laboratory

Deployable Composite Toroid Holding a Circular Prestressed Membrane

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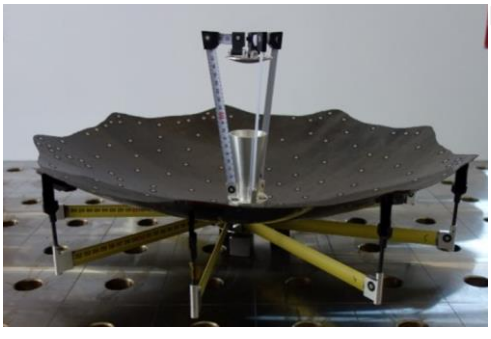
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Deployable space structures with tape springs

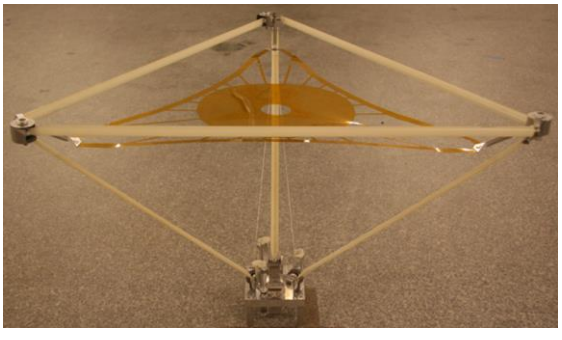
Numerous examples of deployable architectures use conventional tape springs:

- Cube-sat size reflector antenna
- Optic support structure



[1]

Complex system with numerous actuations



[2]

Loss of aperture due to triangular shape

An unexplored solution for deployment → curved tape spring

[1] L. Dufour, et al. "Origami Deployable Reflector Antenna for CubeSats," in AIAA Scitech 2021
[2] J. N. Footdale, et al. "Design and Testing of Self-Deploying Membrane Optic Support Structure Using Rollable Composite Tape Springs," SSDM 2013

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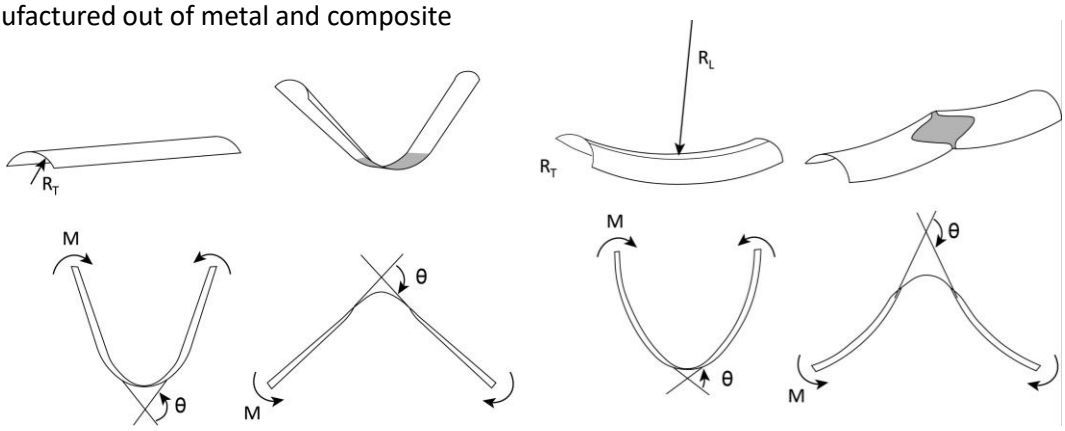
Straight vs Curved tape springs

Conventional tape spring:

- Single transverse curvature
- Stable and uniform equal and opposite-sense folding
- Manufactured out of metal and composite

Curved tape spring:

- Both transverse and longitudinal curvature
- Non-developable surface
- Only manufactured out of metal



[3]

[3] K. A. Seffen, Z. You, and S. Pellegrino, "Folding and deployment of curved tape springs," *International Journal of Mechanical Sciences*

3

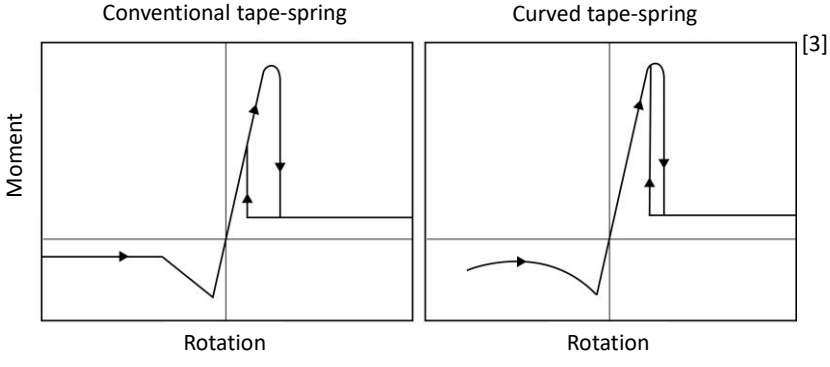
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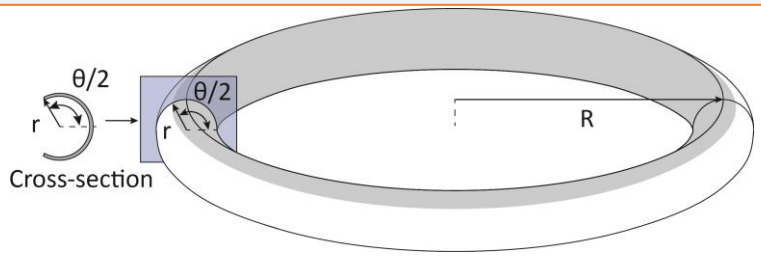
[3]

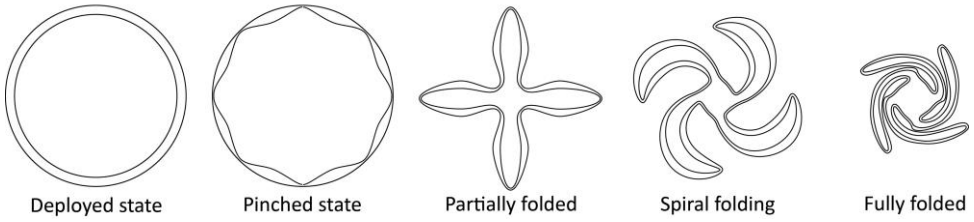
[3] K. A. Seffen, Z. You, and S. Pellegrino, "Folding and deployment of curved tape springs," *International Journal of Mechanical Sciences*

4

Folding scheme of novel deployable architecture

- Extend curved tape-spring to full circular structure:
  → Inner surface of toroidal shell
- Leverage the stiff continuous edge
- Apply pinching and folding of curved tape-spring to fold





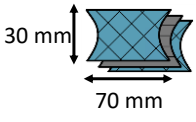
Objectives:
- Manufacture composite toroidal shell
- Identify suitable attachment method
- Demonstrate folding and deployment

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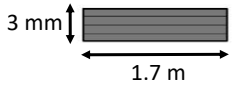
Manufacturing of composite toroidal shell

Toroidal shell requires discretization of composite material

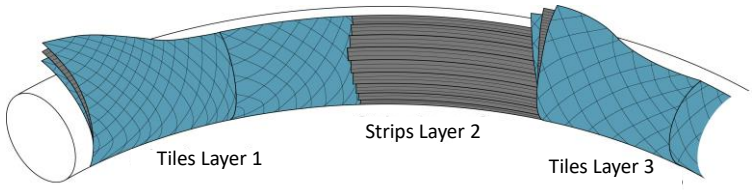
7-plies: [±45°,0°,±45°,0°, ±45°,0°,±45°]



- 3-lamina hybrid stack
- Curved tile
- 67 µm thick



- 1-lamina CF layer
- Rectangular tile
- 17 µm thick





Composite layup on mold



Peel-ply layer



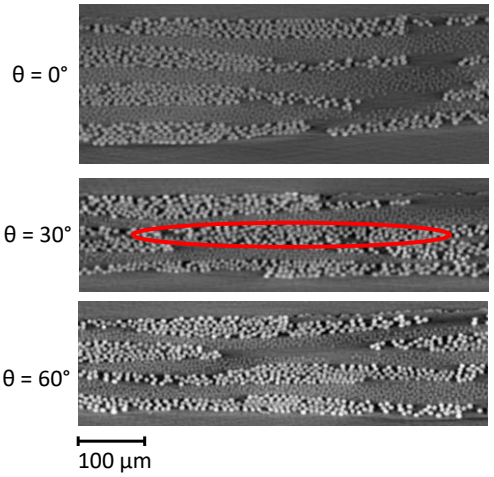
Breather layer



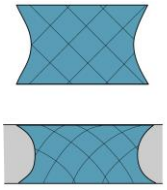
Vacuum bag

μ-CT measurements of composite toroidal shell

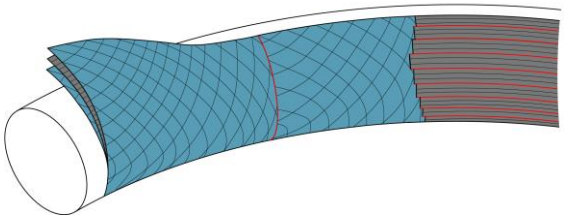
Effects of draping and discretization to composite's microstructure

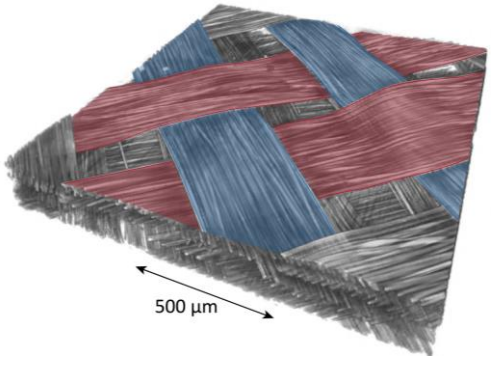


Draping



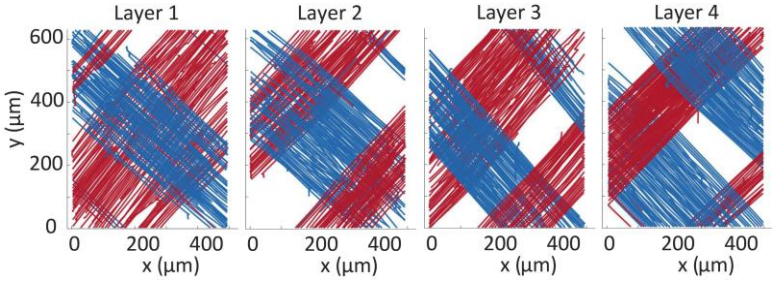
Discontinuities

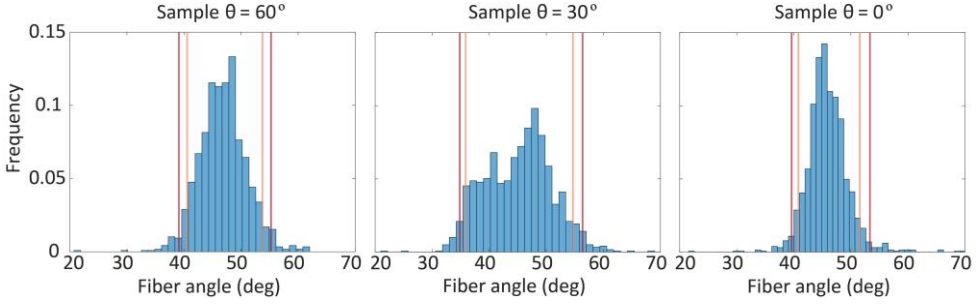




μ-CT measurements of composite toroidal shell

- Fiber tows identification through image processing algorithm
- Measurement of fiber angles and distribution between samples
- Results indicate gap in middle 0° CF layer caused greater fiber dispersion





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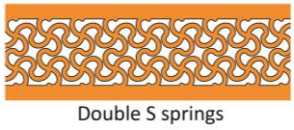
Integration of toroidal shell to prestressed membrane

Functional membrane needs to be:

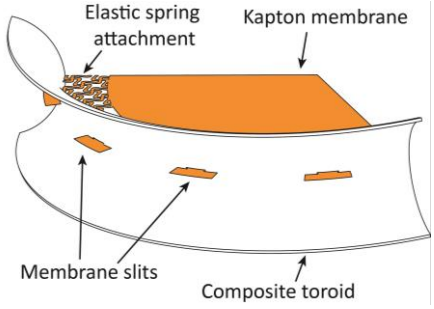
- Prestressed
- Connected to shell

Double s spring design

- Highly compliant
- Allowing high-strains without plasticity



Double S springs





Top view

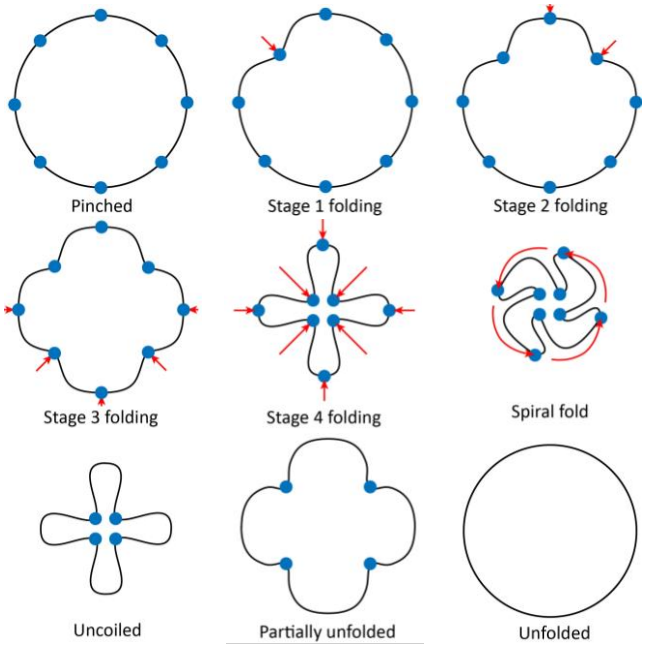


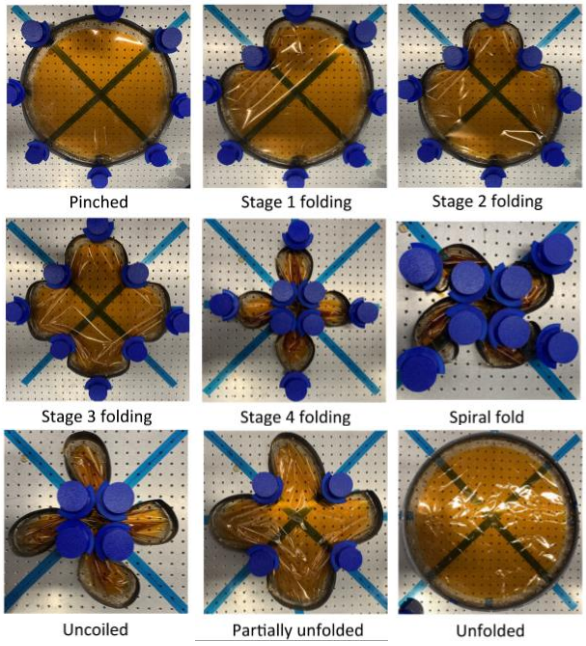
Isometric view

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9

Folding and deployment experiment





10

Conclusion and future work

Novel deployment scheme for high-strain composite structure:

- Advantage of stiff pretensioned circular perimeter
- Maximum functional area to perimeter mass ratio
- Small stowed volume through foldable curved tape-spring

Investigation demonstrated:

- Manufacturing feasibility with thin hybrid composite
- Prestressing through compliant thin-film attachment

Future work:

- Investigate shape accuracy of functional membrane
- Apply and scale technology to spacecrafts



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Acknowledgments

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Thank you for listening

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